
Exercises

Advanced Lesson 3

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Your Math Class: MATH 7013 - Advanced Mathematical Modeling

Note: Most of the following exercises are expressed in every day mathematical notation rather than proper *Mathematica* notation. You MUST convert every day notation into *Mathematica* notation for your code to work properly. For example:

Equations must be changed from "=" to "==" to work correctly (i.e. $x^2 == 2x$ not $x^2 = 2x$).

Functions must be expressed as $f[x]$ and not $f(x)$ to work correctly.

Functions must be defined using "_" to work correctly (i.e. $f[x_] = 2x^2 + 1$ and not $f[x] = 2x^2 + 1$).

Mathematica functions must start with a capital (i.e. $\text{Sin}[x]$ not $\text{sin}[x]$).

Your functions should start with a lower case letter (i.e. $f[x]$ not $F[x]$).

Special symbols must be entered correctly (i.e. e from the palette and not e).

These exercises begin your training in making this conversion on your own automatically.

Remember to clear all variables that you will use below: **Clear[a, b, c, x, y, f, n]**

Clear[a, b, c, x, y, f, n];

1: Part 1: Solve: **$ax^4 + bx^2 + c = 0$** for: **x**

Part 2: Solve: **$ax^4 + bx^2 + c = 0$** for: **a**

Solve[a x⁴ + b x² + c == 0, x]

Solve[a x⁴ + b x² + c == 0, a]

2: Part 1: Solve: $a x^2 + b y + c = z$ and $y + a z^2 + b x^2 + c = 1$ for: x and y
Simplify the answer.

Did **Simplify** shorten the answer in this case? (Put your words into a Text Cell.)

Part 2: Solve: $a x^2 + b y + c = z$ and $y + a z^2 + b x^2 + c = 1$ for: a and x

```
sol2a = Solve[{a x^2 + b y + c == z, y + a z^2 + b x^2 + c == 1}, {x, y}]
```

```
Simplify[sol2a]
```

```
sol2b = Solve[{a x^2 + b y + c == z, y + a z^2 + b x^2 + c == 1}, {a, x}]
```

No. Simplify changed almost nothing here (140 parts before, 138 after). The solutions are already about as short as Mathematica can write them.

3: Part 1: Generate a plot of $e^{2x} \sin(3x) \cos(4x)$ for $0 \leq x \leq 2$ (Remember to enter e from the palette and not e .)

Part 2: Evaluate: $\int e^{2x} \sin(3x) \cos(4x) dx$

Part 3: **Simplify** the answer.

Did **Simplify** shorten the answer in this case? (Put your words into a Text Cell.)

```
Plot[e^{2 x} Sin[3 x] Cos[4 x], {x, 0, 2}]
```

```
int3 = \int e^{2 x} Sin[3 x] Cos[4 x] dx
```

```
Simplify[int3]
```

No. The integral comes back the same size after Simplify, because it is already in its shortest form.

4: Part 1: Generate a plot of $e^{2x} \sin(3x) \cos(4x)$ for $0 \leq x \leq 2$

Part 2: Differentiate: $e^{2x} \sin(3x) \cos(4x)$ with respect to x .

Part 3: **Simplify** the answer. Did **Simplify** shorten the answer in this case?

Part 4: Define a function $f(x)$ to be $e^{2x} \sin(3x) \cos(4x)$. Remember to use $f[x_] = \dots$ and not $f[x]$.

Hint: Copy, paste and edit your work from Parts 1, 2 and 3 to simplify your work in the following Parts.

Part 5: Generate a plot of $f(x)$ for $0 \leq x \leq 2$

Part 6: Differentiate: $f(x)$ with respect to x using the $f'[x]$ notation.

Part 7: Use **Simplify[f'[x]]** to simplify the answer.

Part 8: Evaluate both $f'(x)$ and $\partial_x(f(x))$ and verify that you get the same result.

Part 9: Evaluate both $f''(x)$ and $\partial_{x,x}(f(x))$ and verify that you get the same result.

Part 10: Evaluate both $f'''(x)$ and $\partial_{x,x,x}(f(x))$ and verify that you get the same result.

Hint: To get $\partial_{x,x,x}f[x]$ edit $\partial_{x,x}f[x]$ by adding $,x$ to get $\partial_{x,x,x}f[x]$

Remember to put your words into one or more Text Cells.

```
Plot[e^{2 x} Sin[3 x] Cos[4 x], {x, 0, 2}]
```

```
d4 = \partial_x (e^{2 x} Sin[3 x] Cos[4 x])
```

```

Simplify[d4]

Clear[f, x];
f[x_] = e2 x Sin[3 x] Cos[4 x];

Plot[f[x], {x, 0, 2}]

f'[x]

Simplify[f'[x]]

{Simplify[f'[x] -  $\partial_x f[x]$ ]}

{Simplify[f''[x] -  $\partial_{\{x,2\}} f[x]$ ]}

{Simplify[f(3)[x] -  $\partial_{\{x,3\}} f[x]$ ]}

```

Simplify does help here: the derivative goes from 46 parts down to 28, because the product rule leaves terms that combine. Parts 8 to 10 show that $f'[x]$ and the partial derivative notation are two ways of writing the same thing: subtracting the two and simplifying gives 0 for the first, second and third derivatives.

5: Find the local (not global) maximum for the function $x^3 - x + 1$

Part 1: Start by defining a function $f(x)$ to be $x^3 - x + 1$.

Part 2: Evaluate both $f'(x)$ and $\partial_x(f(x))$ and verify that you get the same result.

Part 3: Evaluate both $f''(x)$ and $\partial_{x,x}(f(x))$ and verify that you get the same result.

Part 4: Plot the function $f(x)$ on the same graph as its derivative $f'(x)$.

Part 5: Find the critical points for $f(x)$ by solving $f'(x) = 0$ for x and then determine the local maximum by checking f'' at the critical points.

Remember to put your words into one or more Text Cells.

```

Clear[f, x];
f[x_] = x3 - x + 1;

{f'[x],  $\partial_x f[x]$ , Simplify[f'[x] -  $\partial_x f[x]$ ]}

{f''[x],  $\partial_{\{x,2\}} f[x]$ , Simplify[f''[x] -  $\partial_{\{x,2\}} f[x]$ ]}

Plot[{f[x], f'[x]}, {x, -2, 2}, PlotLegends -> {"f", "f'"}]

crit = Solve[f'[x] == 0, x]

Table[{x /. s, f''[x] /. s, f[x] /. s}, {s, crit}]

```

The critical points are $x = -1/\sqrt{3}$ and $x = +1/\sqrt{3}$. The second derivative is $-2\sqrt{3}$ at the first, which is negative, so that is the local maximum; at the second it is $+2\sqrt{3}$, a local minimum. The plot agrees: the curve turns over on the left and turns back up on the right.

6: Part 1: Simplify $\text{Cos}[n \pi]$ without assumptions:

Part 2: Simplify $\text{Cos}[n \pi]$ assuming that $n \in \text{Integers}$:

Part 3: Simplify $\text{Cos}[n \pi]$ assuming that n is even, i.e. $n/2 \in \text{Integers}$:

Part 4: Simplify $\text{Cos}[n \pi]$ assuming that n is odd:

Clear[n];

Simplify[Cos[n π]]

Simplify[Cos[n π], n $\in \mathbb{Z}$]

Simplify[Cos[n π], $\frac{n}{2} \in \mathbb{Z}$]

Simplify[Cos[n π], $\frac{n-1}{2} \in \mathbb{Z}$]

7: Part 1: Generate a plot of $x^5 \sin(x)$ for $-1 \leq x \leq 1$

Part 2: Evaluate the following indefinite integral to obtain the generic answer:

$$\int x^5 \sin(x) dx$$

Part 3: Evaluate the following definite integral:

$$\int_2^3 x^5 \sin[x] dx$$

Remember to put your words into a Text Cell.

Plot[x⁵ Sin[x], {x, -1, 1}]

$$\int x^5 \sin[x] dx$$

$$\int_2^3 x^5 \sin[x] dx$$

The indefinite integral is what repeated integration by parts gives. The definite integral from 2 to 3 comes out exactly as $112 \cos[2] - 63 \cos[3] + 40 \sin[2] - 15 \sin[3]$, which is about 50.016.

8: Part 1: Try to generate a plot of $\sqrt{x}$ for $-1 \leq x \leq 1$ and note what happens for $-1 \leq x < 0$.

You will better see the plot if you use the plot option: **PlotStyle $\rightarrow$ Thickness[0.01]**

Plot[..., PlotStyle $\rightarrow$ Thickness[0.01]]

Part 2: Evaluate the following indefinite integral to obtain the generic answer:

$$\int \sqrt{x} dx$$

Part 3: Evaluate the following definite integral:

$$\int_1^2 \sqrt{x} dx$$

Part 4: Evaluate the following definite integral:

$$\int_a^b \sqrt{x} dx$$

Why is it not surprising that the answer depends upon the values of both **a** and **b**?

Part 5: Evaluate the following definite integral **assuming** that **0 < a < b**:

$$\int_a^b \sqrt{x} dx$$

Why is it not surprising that the answer is both convergent and a real number?

Part 6: Evaluate the following definite integral **assuming** that **a < 0 < b**:

$$\int_a^b \sqrt{x} dx$$

Why is it not surprising that the answer is both convergent and not a real number?

Plot[$\sqrt{x}$, {x, -1, 1}, PlotStyle $\rightarrow$ Thickness[0.01]]

$$\int \sqrt{x} \, dx$$

$$\int_1^2 \sqrt{x} \, dx$$

$$\int_a^b \sqrt{x} \, dx$$

`Integrate[ $\sqrt{x}$ , {x, a, b}, Assumptions  $\rightarrow 0 < a < b$ ]`

`Integrate[ $\sqrt{x}$ , {x, a, b}, Assumptions  $\rightarrow a < 0 < b$ ]`

Sqrt[x] is not a real number for negative x, so on $-1 \leq x < 0$ there is nothing to draw and the curve only begins at 0. The indefinite integral is $(2/3) x^{3/2}$, and from 1 to 2 it is $(2/3)(2 \sqrt{2} - 1)$. From a to b with no assumptions Mathematica cannot tell whether the interval crosses the negative axis, so it answers with a ConditionalExpression stating when its result is valid. Assuming $0 < a < b$ the whole interval is where Sqrt[x] is real, so the answer is real. Assuming $a < 0 < b$ part of the interval is where Sqrt[x] is imaginary, so the answer picks up a factor of i .